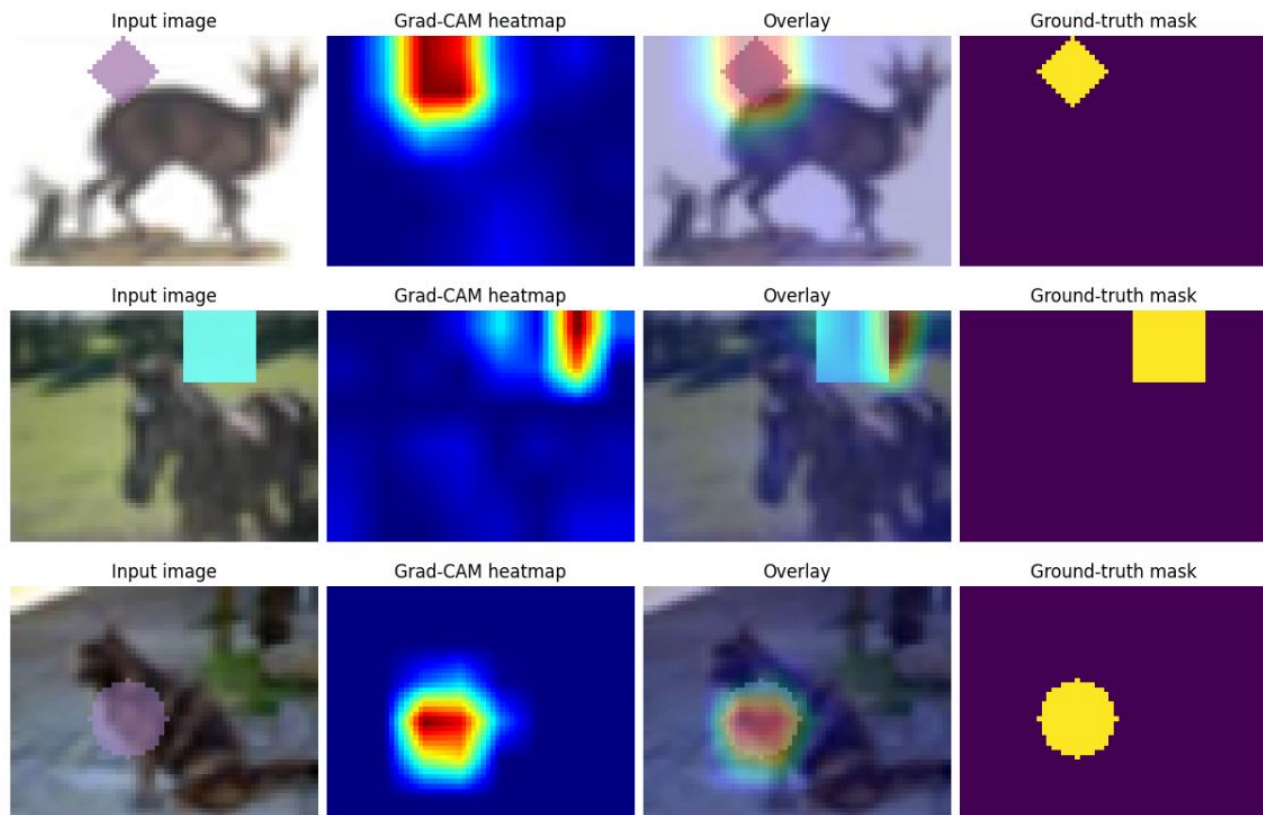


Project Report for Homework 2: Image segmentation without user input

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- An exemplary visualization of the output of the Grad-CAM



- A concise description of each approach for the SAM pipelines

In the segmentation task I designed a pipeline (inside classes `ForeWithBack` and `ForeNoBack` classes) that takes only an image and outputs a segmentation of the certain shape:

(Model + Target Layers +) Input Image -> Grad-CAM -> Heatmap -> Coordinates extraction -> SAM -> Predicted Mask

The input to SAM should be the image and coordinates of points that are likely to be inside or likely to be outside. For calculating those points I tried two different approaches:

- *At least one foreground point, without background points (ForeNoBack class):*
In this approach, I calculated only one foreground point for each image by finding the index of the maximum value in the Grad-CAM heatmap. I then passed this single coordinate to the SAM predictor to generate the final mask.

- *At least one foreground point and at least one background point (ForeWithBack class):*

For this approach, I calculated one foreground point from the Grad-CAM heatmap's maximum and one background point from its minimum for each image. I then used both coordinates as input for the SAM predictor to refine the segmentation output.

- A presentation of all metrics:

```
Hit rate for ForeNoBack class = 0.686
Hit rate for ForeWithBack class = 0.686
Distance metric for ForeNoBack class = 5.9822886824597985
Distance metric for ForeWithBack class = 5.9822886824597985
IoU for ForeNoBack class = 0.727649423360838
IoU for ForeWithBack class = 0.7210771453191451
```

- *Pipeline Metric:*

IoU = Area of Overlap / Area Of Union. Firstly, I calculated mean IoU for each batch and then took an average of them.

- *Metrics for generated points:*

- Hit rate : how often foreground points were inside the ground-truth mask
Firstly, I calculated mean hit rate for each batch and then took an average of them.
- Distance: distance from the foreground points to the center of mass of the ground-truth mask. Firstly, I calculated mean distance for each batch and then took an average of them.

As we can see on the image, all the metrics beside IoU are the same for all approaches. Hit rate shows us that in most cases foreground points fell inside the ground-truth mask. Distance metric shows us that the average distance between each foreground point and center of mass of the ground-truth mask for each image is approximately 6 pixels. IoU for both approaches is almost the same, but for the first approach without background points is a little bit higher/better.

- Discussion on potential areas for improvements:

I calculated only one foreground(and background) point for each image, but we can try to find more foreground(and background) points (by taking few neighbors of already found points e.g.) to improve metrics. Moreover, we can try to change target layers for Grad-CAM.